The Sizland Land Acquisition Workflow

A proprietary P2P/Escrow interface that bridges blockchain logic with real-world legal milestones for land purchases. This system implements a "Notary-Mediated" workflow where the Legal Notary acts as the Oracle for fund releases.

System Architecture

Core Components

1. **MultiSigWallet.sol** - Base multi-signature wallet implementation
2. **NotaryMediatedEscrow.sol** - Main escrow contract with milestone-based releases
3. **EscrowFactory.sol** - Factory for modular escrow deployment
4. **Frontend Interface** - React-based 4-phase workflow UI
5. **Document Hashing System** - IPFS + on-chain verification

Key Features

- **Multi-Sig Security**: Requires multiple confirmations for fund releases
- **Notary Oracle**: Legal Notary verifies legal milestones before fund releases
- **Document Hashing**: All legal documents hashed on-chain for verification
- **4-Phase Workflow**: Discovery → Safety Gate → Execution → Final Settlement
- **Escrow Factory**: Modular deployment for any land purchase

Quick Start

Prerequisites

- Node.js >= 16.0.0
- Hardhat development environment
- MetaMask or compatible Web3 wallet

Installation

```
# Clone the repository
git clone <repository-url>
cd multisig

# Install dependencies
npm install

# Compile contracts
npx hardhat compile

# Run tests
npx hardhat test

# Deploy contracts
npx hardhat run scripts/deploy.js --network localhost
```

Frontend Setup

```
# Navigate to frontend directory
cd frontend

# Install dependencies
npm install
```

```
# Start development server
npm start
```

4-Phase Workflow

Phase 1: Discovery & Lock-up

- **User Action**: Client deposits total budget (e.g., $5,000) into Sizland Treasury
- **System Action**: Funds are locked in multi-sig
- **UI Component**: Curated list of "Digital Proofs" (Geo-tagged photos/videos)

Phase 2: Safety Gate ($200 Due Diligence)

- **Condition**: First milestone of $200 unlocked for plot selection
- **Backend Logic**: Goes to Doddy's wallet for Registry Search and site visit
- **UI Component**: "Document Vault" for viewing Registry Search and Site Visit Report

Phase 3: Execution (Commitment Milestone)

- **Condition**: Both parties sign draft agreement on-platform
- **On-Platform Signing**: Client or seller uploads draft, both parties sign with real-time notifications
- **Notary Role**: Doddy verifies signed documents and clicks "Proceed to Next Step"
- **Smart Contract Action**: Release percentage-based deposit to seller after notarization
- **UI Component**: Document signing system with party status tracking

Phase 4: Final Settlement & Notary Release

- **Condition**: Doddy confirms Original Title and LCB Consents
- **Smart Contract Action**: Release balance to seller + facilitation fee to Sizland
- **UI Component**: "Finalization Status" showing transfer progress

Smart Contract Details

NotaryMediatedEscrow.sol

Key Functions

```
// Create new escrow
function createEscrow(address _seller, address _notary, uint _totalAmount) returns (uint)

// Fund escrow with total amount
function fundEscrow(uint _escrowId) payable

// Upload draft sale agreement (client or seller)
function uploadDraftAgreement(uint _escrowId, bytes32 _hash)

// Sign document on platform (both parties)
function signDocument(uint _escrowId, DocumentType _docType)

// Notarize signed document and proceed (notary only)
function notarizeSignedDocument(uint _escrowId, DocumentType _docType)

// Confirm notarized agreement (client only)
function confirmNotarizedAgreement(uint _escrowId)

// Upload other documents (notary only)
```

```
function uploadDocument(uint _escrowId, DocumentType _docType, bytes32 _hash)

// Finalize transfer (notary only)
function finalizeTransfer(uint _escrowId)
```

Milestone Structure

```
enum Milestone {
    DueDiligence,    // $200 for registry search and site visit
    Deposit,         // Percentage-based deposit to seller
    FinalSettlement  // Balance + facilitation fee
}
```

Document Types

```
enum DocumentType {
    RegistrySearch,
    SiteVisitReport,
    DraftSaleAgreement,
    SignedSaleAgreement,
    OriginalTitle,
    LCBConsent
}

enum SigningStatus {
    NotStarted,
    Draft,
    ClientSigned,
    SellerSigned,
    BothSigned,
    NotaryVerified
}
```

EscrowFactory.sol

Key Functions

```
// Create escrow with Doddy as default notary
function createEscrowWithDoddy(address _seller, uint _totalAmount) returns (address)

// Create escrow with custom notary
function createEscrow(address _seller, address _notary, uint _totalAmount) returns (address)

// Batch create multiple escrows
function createBatchEscrows(address[] memory _sellers, address[] memory _notaries, uint[] memory _total
```

Frontend Components

Main Interface

- **EscrowInterface.jsx** - Main dashboard with phase navigation
- **ProgressBar.jsx** - Real-time progress tracking
- **MultiSigVerification.jsx** - Transaction confirmation interface

Phase Components

- **DiscoveryPhase.jsx** - Property selection and fund locking

- **SafetyGatePhase.jsx** - Due diligence document verification
- **ExecutionPhase.jsx** - Sale agreement confirmation and deposit release
- **FinalSettlementPhase.jsx** - Title verification and final settlement

Supporting Systems

- **DocumentHashingSystem.jsx** - IPFS upload and on-chain hashing
- **App.jsx** - Main application component with wallet integration

Security Features

Multi-Sig Verification

- All fund releases require multiple confirmations
- Configurable number of required signatures
- Transaction queue with status tracking

Document Hashing

- SHA-256 hashing of all legal documents
- On-chain storage of document hashes
- IPFS storage for original documents
- Public verification capability

Condition Hooks

- Progress bar with real-time status updates
- Multi-sig verification for fund releases
- Document hashing verification before payments
- State-based access control

Financial Flow

Fee Structure

```
Total Purchase Amount: $5,000
 Due Diligence Fee: $200 (4%)
 Deposit (10%): $500
 Final Balance: $4,200
 Facilitation Fee: $100 (2% of total)
```

Release Schedule

1. **$200** → Doddy (Due Diligence)
2. **$500** → Seller (Deposit after Sale Agreement)
3. **$4,200** → Seller (Final Balance after Title Transfer)
4. **$100** → Sizland (Facilitation Fee)

Development

Contract Deployment

```
# Start local Hardhat network
npx hardhat node

# Deploy contracts (in separate terminal)
npx hardhat run scripts/deploy.js --network localhost
```

Frontend Development

```
cd frontend
npm install
npm start
```

Testing

```
# Run contract tests
npx hardhat test

# Run frontend tests
cd frontend
npm test
```

Configuration

Environment Variables

```
# Network Configuration
NETWORK_ID=1337
RPC_URL=http://localhost:8545

# Contract Addresses (after deployment)
MULTISIG_WALLET_ADDRESS=0x...
NOTARY_MEDIATED_ESCROW_ADDRESS=0x...
ESCROW_FACTORY_ADDRESS=0x...

# IPFS Configuration
IPFS_GATEWAY=https://ipfs.io/ipfs/

# Notary Configuration
DODDY_NOTARY_ADDRESS=0xDoddyNotary12345678901234567890123456789
SIZLAND_TREASURY_ADDRESS=0xSizlandTreasury123456789012345678901234567
```

Integration Guide

Backend Integration

1. **Deploy contracts** to desired network
2. **Configure EscrowFactory** with Doddy's address
3. **Set up multi-sig owners** (Doddy, Sizland, trusted third party)
4. **Fund Treasury** with initial capital if needed

Frontend Integration

1. **Update contract addresses** in App.jsx
2. **Configure Web3 provider** for target network
3. **Set up IPFS gateway** for document storage
4. **Test wallet integration** with MetaMask

Legal Integration

1. **Onboard Doddy** as authorized notary
2. **Establish document templates** for each phase
3. **Set up verification process** for legal documents
4. **Create dispute resolution** mechanism

Important Notes

Security Considerations

- Always verify contract addresses before deployment
- Test thoroughly on testnet before mainnet deployment
- Ensure proper multi-sig configuration
- Regular security audits recommended

Legal Compliance

- Ensure proper licensing for notary services
- Comply with local land transfer regulations
- Maintain proper audit trails
- Consider jurisdiction-specific requirements

Operational Requirements

- Doddy must have proper notary certification
- Sizland Treasury must be properly secured
- Document verification process must be established
- Customer support infrastructure needed

Support

For technical support or questions:

- Create an issue in the repository
- Contact the development team
- Review documentation and test cases

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